COMPUTER PROCESSOR

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*Introduction

- A processor or Central Processing Unit (CPU) is an electronic circuit that can execute computer programs.
- A processor is the logic circuitry that responds to and processes the basic instructions that drives a computer.
- A computer processor analyzes data and controls data flow in a computer.
- It handles the central management functions of a high-powered PC.

Image of processor

Working of processor

The working of processor mainly consist four steps i.e. Fetch, Decode, Execute and Write back. These steps are discussed below :

- **Fetch :** During the fetch step, the processor retrieves program instructions from memory.
- **Decode :** In this step, the instruction is broken down into parts

- **Execute :** In the execute step, CPU performs the operation implied by the program instructions.
- **Write back :** During the write back step, the Processor writes back the results of execution, to the computer's memory.

Basic knowledge about processor

Clock speed : Clock speed is a measure of how quickly a computer completes basic computations and operations. It is measured as a frequency in hertz. Generally this speed varies 2.4-3.4 GHz.

Core : A processor core is a hardware unit in the processor architecture that can execute instructions sent to it.

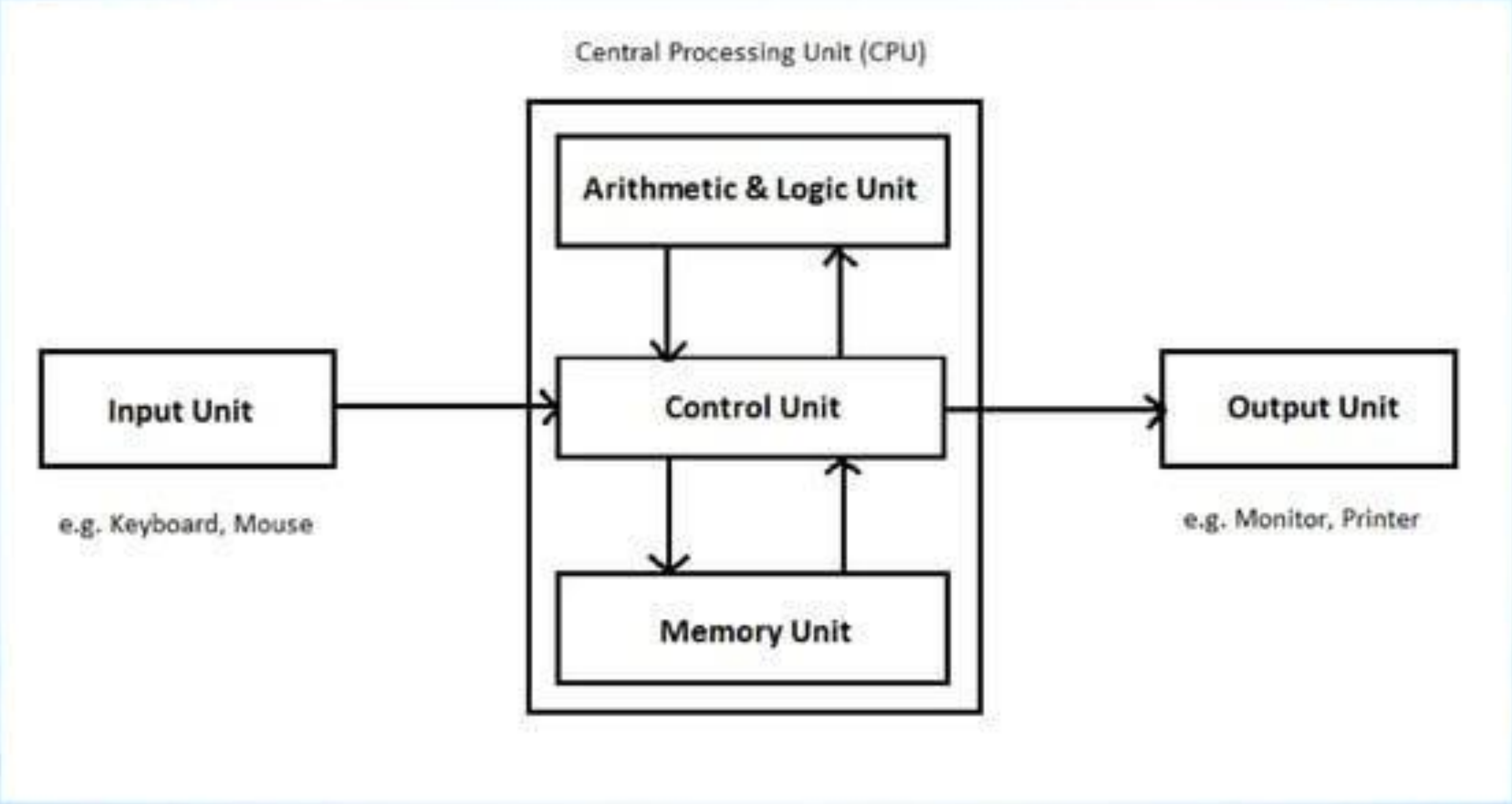
Hyper threading : Thread are the virtual core and work like a real core inside cpu. When cores assembled with multi threads then it is known as Hyper threading technology.

Cache : The cache is the first block of RAM which interact between the main memory and CPU using cache controller chip. This memory helps processor to fetch instructions in quick routine and is very faster than RAM

MULTI PROCESSING : Simultaneous processing with two or more processors in one computer or two or more computers processing together.

PARALLEL PROCESSING : The simultaneous use of more than one cpu or processor core to execute a program or multiple computational threads

Components of CPU



What is single, double and multi core processors?

Single core: Has one core to process different operations like intel Pentium.

Dual core : Has two cores to process operations; able to process more information at the same time compare to single core like intel core i3 and i5.

Quad core: Contains two dual core processors in one integrated circuit and generally used for multi tasking like intel core i7.

Some popular brands of processor

Intel : The Intel computer processor is exclusively designed by Intel. Its latest and most popular models include Intel hyper thread technology that speed up processor speed .

intel

AMD : The AMD computer processor is exclusively made by Advanced Micro Devices, Inc. (AMD). It provides excellent performance and value. It is compatible with most off-the-shelf computer programs and applications. Some AMD Computer Processors are programmed with built-in antivirus protection

AMD

Tomorrow, processor will be more energy efficient than present time.

They have high clock speed like 10 to 15 GHz and have 8 to 10 cores with hyper threading or new equivalent technology.

Their size will be more small than today's processors i.e. nano processor will come in place of micro processor.

They will be more cheaper and reliable than compare to present time.

Conclusion

Multi-core processors represent an important new trend in computer architecture.

Decreased power consumption and heat generation also helpful.

Minimized wire lengths and interconnect latencies.

So this is the way how modern computing is done.

THE END